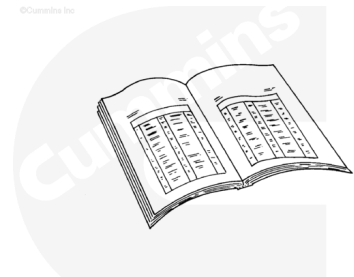


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

Note : The lubricating oil pump does not need to be removed on engines manufactured prior to engine serial number (ESN) 31117702. These engines use a retaining ring to secure the pressure regulator assembly into the lubricating oil pump and can be accessed without removing the lubricating oil pump.

- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-031-tr.html)

Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.

Note : For engines that do not need the oil pump removed, use the pressure regulator removal tool, Part Number 5299024, to remove the pressure regulator.

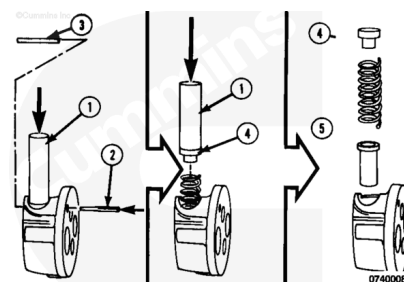
Use an arbor press and mandrel (1) to hold the retainer (4) in place while the dowel pin (3) is being removed.

Some models use a short dowel and roll pin to hold the retainer.

Remove the dowel pin (3) with a brass drift (2).

Slowly raise the arbor press to relieve the spring tension.

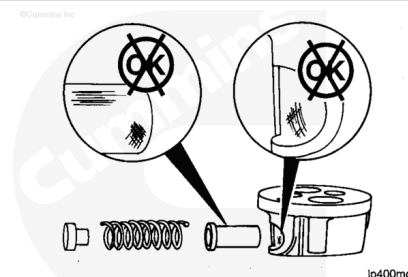
Remove the retainer (4), spring, and regulator plunger (5).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the parts with solvent and dry with compressed air.

Inspect the plunger for damage.

The plunger **must** be replaced, if damaged.

Inspect the plunger bore in the oil pump for damage.

Place the plunger in the bore and check for free movement.

If the bore in the lubricating oil pump is damaged, the lubricating oil pump **must** be reconditioned or replaced. Refer to Procedure 007-031 in Section 7.

(/qs3/pubsys2/xml/en/procedures/20/20-007-031-tr.html)

Check the free length of the spring.

Spring Free Length (Approximately)	
mm	in
95	3.75

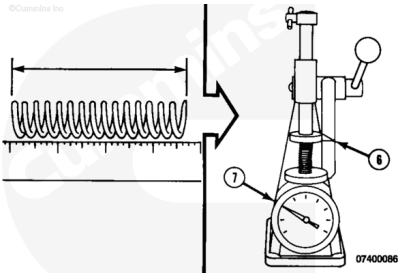
Measure the spring force (7) at the working height (6) with valve spring tester, Part Number 3375182.

Spring Working Height	
mm	in
63.88	2.515

Spring Force

n		lb
300	MIN	68
310	MAX	70

If the spring is **not** within specifications, it **must** be replaced.



Install

Some models use a short dowel and a roll pin. Install the short dowel even with the gasket surface before installing the remaining parts.

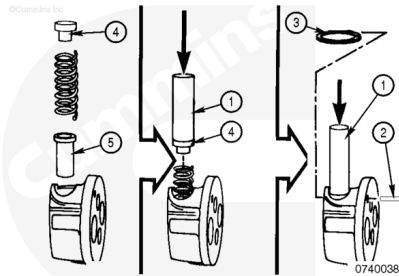
Lubricate the regulator plunger, lubricating oil pump bore with clean engine oil.

Note : For engines that did not need the oil pump removed, use the regulator removal tool, Part Number 5299024, to install the regulator.

Install the plunger (5), spring, and retainer (4).

Use a mandrel (1) and a arbor press to compress the spring.

Use a mallet to install the roll pin (3) even with the casting.



Finishing Steps

- If removed, install the lubricating oil pump. Refer to Procedure 007-031 in Section 7.

(/qs3/pubsys2/xml/en/procedures/20/20-007-031-tr.html)

- Start and operate the engine and check for leaks.



Last Modified: 16-Oct-2014
